



Faculty of Computer Studies and Information Technology
Department of Information Technology
IT 303 / IT 303L – SYSTEM INTEGRATION AND ARCHITECTURE 1
LABORATORY ACTIVITY 3 — First Semester, AY 2026-2027

SHARED VERSION — every student does the same work with the same values. If you meant to issue the individual version, it is in Individual Activities\.

Activity 3: Second part joined, interface note, absence behaviour shown

Intended Learning Outcome

ILO 3: Join a second part to the loop and write the interface note for the junction it creates.

Time allotment: 3 hours — the second laboratory session for this outcome. The first session covers the taught work, so ILO 3's 6 hours in the syllabus is the two sessions together.

Objectives

- Add a second part to the system and identify the junction it creates.
- Debounce a mechanical input and show the difference it makes.
- Write a complete five-field interface note for that junction.
- Observe and record the absence behaviour by disconnecting the other side.

Materials

- Your measured circuit from Activity 2.
- A blank interface note with its five field names already written down.
- Tool for this activity: Tinkercad Circuits.

Procedure

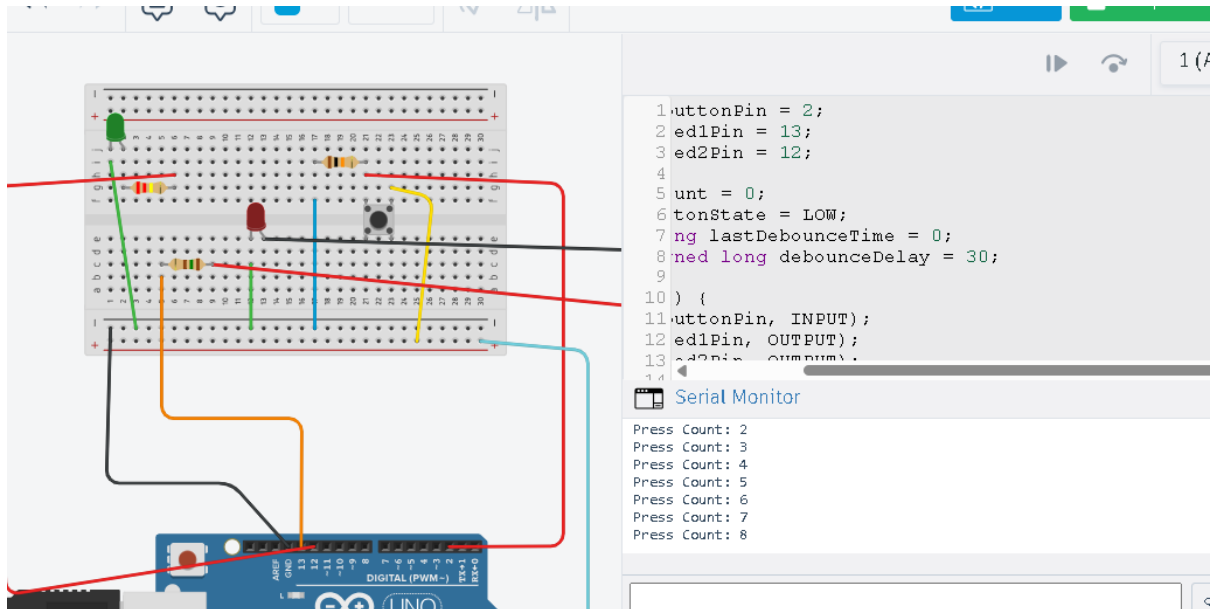
1. Add a second indicator on a different pin, with its own series resistor. This is your new part.
2. Extend the program so the second indicator shows something different from the first — for example, it counts button presses and lights every third one.
3. Print the press count on every press. Press the button once, slowly and deliberately, and read the count.
4. Record what you see. One physical press will usually register as two or three.
5. Fix it: after accepting a change, ignore any further change for 30 milliseconds. Press once more and confirm the count now rises by exactly one.
6. Now write the interface note for the button junction. Fill all five fields from what you have observed, not from what you assume. The timing field must mention the 30 ms.
7. Stop the simulation. Delete the wire between the button and the input pin. Start it again and watch for thirty seconds.
8. Record what the system does. With a pull-down, a disconnected button reads exactly like an unpressed one — so write that in the absence field, honestly.
9. Move the resistor from pull-down to pull-up, restart, and watch again. Record the difference.



10. Restore the working wiring, demonstrate the whole loop, and add the new junction to your integration map.

What to Submit

- The recorded counts, before and after debouncing.



- The complete five-field interface note.

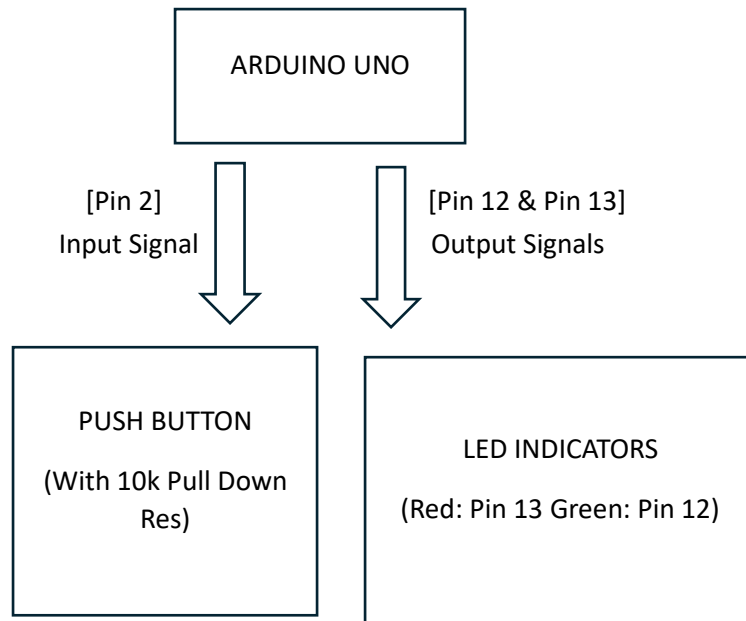
Field Name	Interface Note Details
Signal	Digital Input (Push Button State)
Units	Volts (V) / Logic Level
Range	0 V (LOW) to 5 V (HIGH)
Timing	Active HIGH; 30 ms software debounce delay filtering mechanical contact bounce
Absence Behavior	Pull-down resistor: reads 0 V (LOW / unpressed). Pull-up resistor: reads 5 V (HIGH / pressed)

- What the system did with the wire removed, for BOTH pull-down and pull-up.
Pull-Down Configuration: Removing the wire connected to Pin 2 leaves the input at 0 V (LOW). The system treats the missing wire as an unpressed button.

Pull-Up Configuration: Moving the resistor to 5V and removing the wire holds Pin 2 at 5 V (HIGH). The system treats the missing wire as a continuously pressed button.



- Your updated integration map.



How This Activity Is Marked

Rubric For Laboratory Activities, from the course syllabus. Each criterion is rated Excellent (4), Very Satisfactory (3), Satisfactory (2) or Needs Improvement (1), and that share of its weight is awarded.

Criterion	Weight	What earns full marks here
Working System	30%	The system responds from end to end at the close of the session, and the part added this session works within it.
Program Quality	25%	The program is correct, readable and uses functions and variables appropriately.
Interface Note and Evidence	20%	Every junction made this session has an interface note giving signal, units, range, timing and absence behaviour, and the recorded readings support it.
Explanation	15%	Explains what each part does and why, including any fault encountered.
Timeliness and Compliance with Instructions	10%	Submitted within the session and follows every instruction given.

Reminders

- Everything runs in a browser on the laboratory computers. You buy nothing, and you install nothing.
- Save your own copy of every file. Work that exists only on a laboratory computer is not backed up.
- Evidence means a screenshot showing the thing AND the state you are claiming — the circuit running, the value readable, the serial line visible.
- Every session ends the same way: the system responds from end to end, and you have demonstrated it. A session that ends with a system that does not run is not finished, whatever else was built.
- Every junction you make gets an interface note, written at the bench while you make it. A note written later from memory records what you believe, not what is true.
- Your device code identifies your work. It is an opaque code and it is the only identifier that belongs in a project — never your name or your student number.

